कम्प्युटर र यसको प्रयोग, सूचना तथा सञ्चार प्रविधि सम्बन्धी सामान्य ज्ञान

1.1 Concept

- कम्प्युटरलाई नेपालीमा **सुसाङ्ख्य** यन्त्र भनिन्छ ।
- कम्प्युटर शब्द भन्नाले ल्याटिन भाषाको कम्पयुटेयर (**computare**) शब्दबाट आएको हो । जसको अर्थ हिसाब गर्नु (to calculate or to compute) हो ।
- "**सुसाङ्ख्य**" शब्दको पर्यायवाची शब्द "गणक" हो ➔त्यसैले "सुसाङ्ख्य" शब्दको अर्थ "हिसाब गर्ने एक प्रकारको यन्त्र" भन्ने हुन्छ।

1.2 History of Computer

1.2.1. कम्प्युटर को युग / Era of Computer

Mechanical Era

- Computers of the mechanical era were typically built using mechanical parts such as gears, shafts, levers, etc. made up of wood, stones and bones. Some of them are so simple in construction and efficient in performing tasks that they remain in use nowadays also.

➢ Abacus - 500 BC	➢ **First Calculating Device** ➢ By Chinese, Russian or Egyptian before 2500 to 3000 years ago or around 500 BC
➢ Napier Bone - 1617 AD	➢ Also, known as Napier Rod ➢ Invented by Scottish Mathematician John Napier
➢ Slide Rule – 1620 AD	➢ Also, known as **First Mechanical Calculator** ➢ Invented by English mathematician Willliam Oughterd
➢ Pascaline – 1642 AD	➢ Invented by French mathematician Blaise Pascal ➢ **Odometer** and Pascaline shares common principle
➢ Leibniz Calculating Device – 1671 AD	➢ Also, known as Steeped Reckoner ➢ Invented by German philosopher and mathematician Baron Gottfried
➢ Jacquard Loom – 1802 AD	➢ Invented by French silk weaver Joseph Marie ➢ introducing the concept of programmable automation using **punched cards**
➢ Charles Babbage (1791-1871 AD)	➢ **Father of computer**, ➢ Mathematician at Cambridge University
➢ Difference Engine – 1822 AD	➢ Invented by Charles Baggage
➢ Analytical Engine – 1833 AD	➢ Invented by Charles Baggage ➢ **First General Mechanical Computer**
➢ Lady Augusta Ada Lovelace	➢ Is a **First Computer Programmer** ➢ Also called "Lady ADA"
➢ Boolean Algebra	➢ Irish mathematician George Bool invented **Boolean algebra**.
➢ Tabulating Machine – 1886	➢ Invented by American census statistician Dr. Herman Hollerith. ➢ Also the founder of **IBM**.
➢ John Von Neumann	➢ Father of stored program concept

Electro-Mechanical Era

- Computers of the electro-mechanical era were built using both electronic and mechanical parts such as resistors, capacitors, vacuum tubes, motors, gears, shafts, levers, etc. Some of them are so simple in construction and efficient in performing certain types of special tasks that they remain in use nowadays also.

Mark I (IBM- ASCC) – 1937 AD	➢ Physics Professor of Harvard University Howard H. Aiken. ➢ **First Electro-Mechanical Computer**
Atanasoff Berry Computer (ABC)	➢ Mathematician Professor John Vincent Atanasoff and student Clifford Berry.
John McCarthy-1955	➢ **Father of artificial intelligence** (AI)
The Colossus- 1994 AD	➢ Designed by **Alan M. Turning (father of Computing)** ➢ Build by British mathematician Alan Mathison Neuman.

Electronic Era

The current era of computers is an electronic era where computers are built using electronic components such as resistors, capacitors, and semiconductor devices such as transistors, MOSFET, IC, microprocessors, etc. Computers of this era have become smaller and so powerful that they can store and process huge amounts of data and can execute complex millions and billions of instructions within a fraction of a second. Powerful processing capacity has enabled computers to pose an Artificial Intelligence (AI).

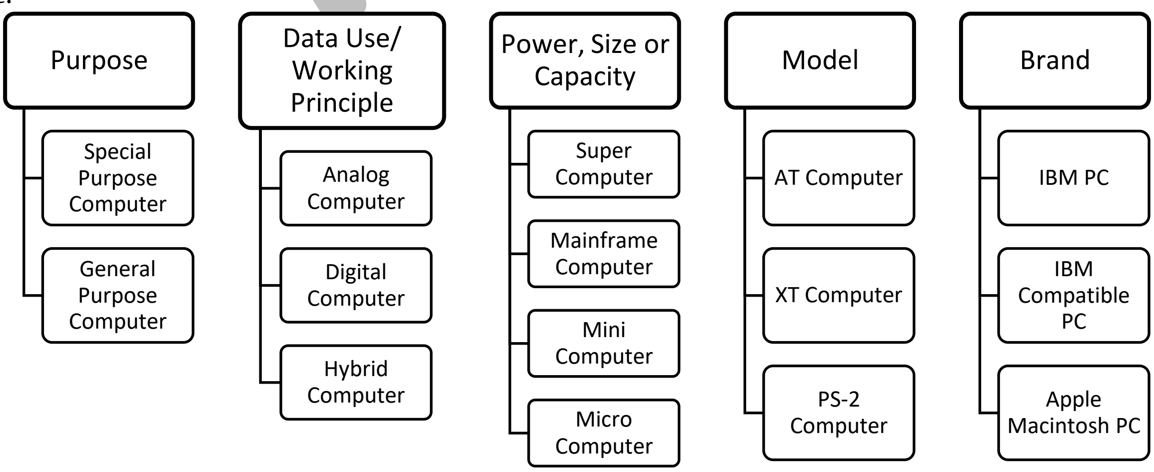
ENIAC -1946 AD	➢ John W.Mauchly & J.Presper Eckest ➢ **First electronic Computer**
EDVAC – 1952 AD	➢ Dr. John Von Neumann, JP Eckert, J.W. Mauchly
EDSAC – 1949 AD	➢ Group of Scientist headed by Professor Maurice Wilkes at Cambridge.
Manchester Mark I	➢ Group of Scientist headed by professor M.H.A Newman ➢ For Practical Use
UNIVAC – 1951 AD	➢ JP Eckert & J.Mauchly ➢ **First** general purpose computer design for **commercial purpose**

1.2.2. Generation of Computer (कम्प्युटरका पुस्ता)

	First Generation	**Second Generation**	**Third Generation**	**Fourth Generation**	**Fifth Generation**
Main Component	Vacuum Tube	Transistors	Integrated Circuit (IC)	Microprocessor	Bio-chips
Technology	Glass Shield Filament	Semiconductor	Large Scale Integration (LSI)	Very Large Scale Integration (VLSI)	Ultra Large Scale Integration (ULSI)
Speed	Milli Second	Micro Second	Nano Second	Pico Second	Fermi / Femto second
Primary Memory	Magnetic Core memory	Magnetic Core memory	Semiconductor memory	Semiconductor memory	Unknown
Secondary Memory	Punched Cards, Paper tapes, Magnetic tape	Magnetic Tapes, Magnetic drums	Magnetic Disk (Floppy Disk, hard disk) and magnetic tape	Magnetic and Optical disk (Hard disk, CD, DVD, Pen Drive)	Unknown
Computer programming Language	1GL (Fully Machine Leve)	2GL (Assembly Language)	3GL (High Level Language)	High Level and 4GL	Natural Language
Example:	UNIAC, ENIAC, EDSAC	IBM 1401, IBM-7094/7030	IBM 360 series IBM 370 series	IBM PC, Apple Macintosh	Unknown
					

1.3 (कम्प्युटरका प्रकारहरू) Types of Computer

There has been an invention of different types of computers. These computers can be classified or grouped into different types based upon their common characteristics such as working principle, purpose, size, capacity, model and brand, etc.



1.3.1 On the basis of purpose or Use (उद्देश्य वा प्रयोगको आधारमा)

a. Special Purpose Computer:	b. General Purpose Computer
• Computers designed for single tasks. • eg: Super Computer, Cash billing Machine, Washing machine, etc	• Computer designed to work in most of the field. • eg: Desktop PC, Laptop, Notebook PC, etc

1.3.2 On the basis of Data type they operate or function or working principle (सञ्चालन गर्ने प्रकारको डाटा वा कार्य वा काम गर्ने सिद्धान्तको आधारमा)

Analog Computer:	Digital computer	Hybrid Computer
1. Computers that are made up of mechanical parts and use analog or naturally occurring continuous data. 2. Analog computers are of Special purpose. 3. eg: Thermometer, Speedometer, and ammeter	1. Computers that are made up of only electronic parts and use discrete or digital dis-continuous data. 2. Digital computers are of General purpose. 3. eg: IBM PC, Macintosh Computer	1. Computers that are made up of a mix of both mechanical and electronic parts, which collect data in analog form but convert to digital form using Analog to Digital Converter (ADC) to process digitally and use Digital to Analog Converter (DAC) if required to provide results in analog form. 2. Are also, known as mix of both analog and digital computer. 3. Hybrid computers are of Special purpose. 4. eg: computer used in radar communication, automated industries, ICU, CT scan, etc.

1.3.3 On the basis of Size or Computing Power or Types of Digital Computer

a. Super Computer:

Most powerful and fastest computer with parallel processing and supports multi-user. Used in scientific research, weather forecasting, etc.

b. Mainframe Computer:

This computer is less powerful than Super computer but more than Minicomputer and supports multi- user with parallel processing. Used as server in big organization, e-mail, web hosting, etc.

c. Mini Computer:

This computer is less powerful than a Mainframe computer but more powerful than Micro Computer and supports multi-user with parallel processing. Used as a server in mid-sized organizations, industries, etc.

d. Micro Computer

This type of computer is designed for a single user so, is also called a Personal Computer (PC). This computer is less powerful than other types and uses a microprocessor as a CPU and supports a single user at a time. There are various types of microcomputers designed as per requirement and some of them are:

Types of Microcomputers:

i. **Desktop Computer:** The computers that are designed to be operated by placing them on a table. They are not easily portable and require to be connected to a power socket for a power supply.

ii. **Workstation:** Most powerful micro-computer similar to a desktop, especially used for heavy processing tasks like graphics designing, animation, etc.

iii. **Laptop Computer:** The briefcase-sized computers that are specially designed to be operated while placed on our lap, so they can be portable and used anywhere. They have a fixed keyboard and a touchpad as a pointing device. It also contains an in-built rechargeable battery for power supply.

iv. **Notebook PC:** The computer is similar to a laptop computer but is compact and small in design.

v. **Palmtop or Handheld:** A small-sized computer that can be used by placing on our hand. Especially contains a rechargeable battery with a backup of more than 24 hours and a touch-screen for major input-output activity. A smartphone is an example of this type.

vi. **Tablet:** This a computer with a screen size and processing power slightly bigger than a Handheld device but with a similar interface.

vii. **PDA- Personal Digital Assistant:** It is small small-sized computer designed for a special purpose to ease some sorts of daily life activities. E.g.: Smart Watch, Calculator, Kindle, Digital diary, etc.

1.3.4 On the Basis of the Model

a. XT Computer: XT stands for Extra or Extended Technology.
   It is old technology with a very slow processing speed.
   It supports only CUI (Command or Character User Interface).
b. AT Computer: It stands for Advanced Technology.
   It is faster in processing speed.
   It supports both CUI and GUI (Graphical User Interface).
c. PS/2 Computer: It stands for personal system-2.
   It is an upgraded version of AT, which is more powerful and faster than the AT computer.

1.3.5 On the Basis of Brand

a. IBM PC

- The personal computers manufactured by International Business Machines (IBM) are known as IBM PCs. IBM is the pioneer in the field of researching and manufacturing computers.

b. IBM Compatible PC

- IBM-compatible PC refers to a class of computers that are technically similar to and uses the technology of the original IBM PC. The computers manufactured by different companies like Acer, HP, Dell, etc. are known as IBM-compatible PC.

c. Apple/Macintosh Computer

- Apple has its own architecture used in its different series of computers, different from IBM.

1.4 कमप्यूटरका आधारभूत शब्दहरु (Terms Used in Computer)

- **डाटा** (**Data**) : डाटा भनेको कच्चा तथ्य वा परिमाणहरूको सङ्कलन हो । Data is the collection of raw facts or figures. Data is represented with the help of characters such as alphabets (A-Z, a-z), digits (0-9) or symbols (+, -, /, *, =).
- For Example: 'Ram' is fact and '13' is a figure.
- **निर्देशन (Instruction) :** निर्देशन भनेको कम्प्यूटरलाई निश्चित कार्य गर्न को लागी भनेर दिइएको आदेश हो। It is command given to the computer to perform a certain task.
- **इन्फरमेशन** (**Information**): Information लाई नेपालीमा सूचना वा जानकारी पनि भनिन्छ। Information is a meaningful result obtained after processing data according to provided instructions. Information are helpful in making decisions and taking actions.
  For Example:
  Ram is 13 years old.
  Ram lives in ward no. 13 of Katmandu Metropolitan City.
- **Data Processing :** उपलब्ध गराइएका निर्देशनहरूको सहायताले डाटालाई सूचनामा रूपान्तरण गर्ने विधि वा प्रक्रियालाई डाटा प्रोसेसिङ (डाटा प्रशोधन) भनिन्छ। The process of converting data into information with the help of provided instructions is known as data processing.
- **प्रोग्राम** (**Progaramme**): कुनै विशेष कार्य गर्न भनेर दिइएको निर्देशनहरूको क्रमबद्ध सेटलाई प्रोग्राम भनिन्छ। (Programme is sequenced set of instruction to perform a certain task.)
- **Files:** A file is an object on a computer that contains data, information, settings, instructions or commands. There are two types of files:
  - Data File: File that contains data or information. Examples: image files, documents, workbooks, music files, video files, etc.
  - Program File: File that contains setting, commands or instruction. Examples: executable file, command file, batch file, etc.

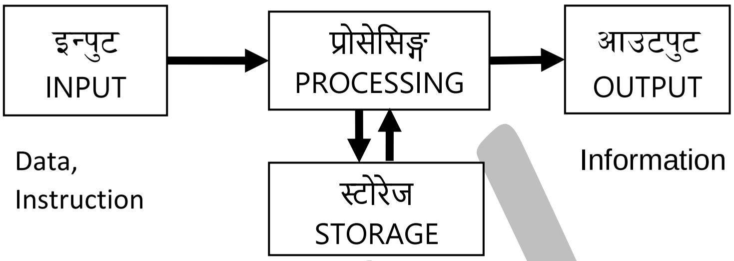
  Some of the different types of data files with their extension are:
  - Image - .jpg, .jpeg, .img, .png, heif, etc
  - Audio - .mp3, .wma, .amr, etc
  - Video – .mp4, .mpeg, .flv, hevf, etc
  - Text - .txt
  - Others - .pdf, .tiff, .csv, .dat, .bmp, etc

  Program files are executable file, command file and batch file and their extension are: .exe, .com, .bat

- **प्रोग्रामिङ ल्याङग्वे** (programming language) : कम्प्यूटरमा प्रोग्राम लेख्न प्रयोग हुने कृत्रिम ल्याङग्वेजलाई प्रोग्रामिङ ल्याङग्वेज भनिन्छ। (is an artificial language used to write a computer program).
- **प्रोग्रामिङ** (**Programming**) : It is the process of developing a computer program. The steps involved in developing a computer program are designing, writing(coding), testing, debugging and maintaining the source code and implementation of computer programs.
- **हार्डवेयर** (**Hardware**): हार्डवेयर भनेको कम्प्युटरमा जडित ,हामीले देख्न, छुन र महसुस् गर्न सकिने सम्पूर्ण भैतिक, मूर्त, इलेक्ट्रोनिक र यान्त्रिक उपकरणहरू हो। (All the physical, mechanical, tangible or electronics parts of a computer that can be touched, seen and felt.)
- **सफ्टवेयर** (**Software**): सफ्टवेयर भनेको कम्प्युटरको गैर-भौतिक भागहरू हुन्, जुन प्रोग्रामहरुको सङ्कलनले बनेको हुन्छ, र जसले कम्प्युटर हार्डवेयर लाई काम गराउँछ। Software is the non-physical parts of a computer, made up of a collection of programs, that makes the computer hardware work.
- **फर्मवेयर (Firmware)**: Firmware is the software or a program stored in a ROM chip which can perform as a hardware. Firmware is stored in a ROM chip at the time of manufacture of computer motherboard. It mainly operates the computer hardware to start up computer.
- **Liveware**: Liveware refers to any person using a computer or simply a computer user. They can be the computer programmers, system analysts, operating staffs, database administrators, hardware technicians and other personnel working in the computer system.
- **Computer System**: The computer along with various hardware units and software that make the computer function and perform the different tasks is collectively known as a computer system. A complete computer system consists of four main components which are Computer Hardware, Computer Software, Data and Computer Liveware (User).
- **Malware**: Malware, short for "malicious software," refers to any intrusive software developed by cybercriminals (often called "hackers") to steal data and damage or destroy computers and computer systems. Some examples of malware are viruses, worms, trojan viruses, spyware, adware, and ransomware.
  - **Computer Virus:** A type of malicious software or program that can replicate itself, spread to other computers by attaching itself to other computer files and cause damage to different types of computer files and systems. Some of its common types are: Boot Sector Virus also known as System Infector Virus, File Infector Virus, Polymorphic virus, Stealth Virus, Multipartite Virus, Resident Virus, Macro Virus, Message Carrying Virus, Browser Hijacker, Web Scripting Virus, etc
  - **Computer Worms:** A malicious software program that can replicate itself and spread to other computers through network connections or external storage device. It spread itself, relying on security failures on the target computer to access it. It will use this machine as a host to scan and infect other computers.
  - **Trojans:** A malicious software program that is disguised as legitimate software and can gain access to a computer system.
  - **Spyware:** A malicious software program that collects information about a computer user without their knowledge.
  - **Adware:** Adware is unwanted software designed to throw advertisements up on your screen, most often within a web browser. This doesn't affect the computer resources but irritates the user.
  - **Ransomware:** A malicious software program that encrypts a victim's files and demands a ransom from the victim to restore access.
- **Peripheral device:** नेपालीमा परि धय उपकरणहरू भनिन्छ । जुन कम्प्युटरको CPU को वरिपरि राखेर चलाइन्छ। उदाहरणका ला ग Keyboard, Mouse, Printer, Monitor, Speaker आदि हुन् ।
- **Server:** Computer को Network वा Internet मा service दिन प्रयोग हुने powerful multi-user computer लाई server computer भनिन्छ। यिनीहरू Web hosting, mail server, data server आदिको रुपमा प्रयोग गरिन्छ।

कम्पयूटरको काम गर्ने सिद्दान्त (Working principle of computer)

- Computer systems work on the basis of input, processing, output, and storage/memory.
- This is also called Information Processing Cycle (IPC).
- Data and instructions are supplied from input devices, then processed and the resulting information is provided from various output devices.
- It stores data, instruction and information on storage devices like hard disk, pen drive, CD, etc. for future use.
- Information is obtained from data processing. A computer can generate, store, process and manage information.



- **Computer**(Current Definition): A computer is an electronic device that processes data, performing calculations and operations based on instructions (programs). It can store, retrieve, and manipulate information in the form of data, which can include numbers, words, pictures, and more. Essentially, a computer takes input, processes it, and provides output, making it a versatile tool for a wide range of tasks.

कम्प्युटर को विशेषताहरू (Characteristics of Computer)

Speed

Computers are capable of performing complex type calculations at very high speed. It can process millions and billions of instructions in a second. The speed of the computer is measured in fractions of a second as Hz, KHz, MHz, GHz, etc.

1 Hertz(Hz)	= 1 cycle per second
1 Kilohertz (KHz)	= 1000 Hz
1 Megahertz(MHz)	= 1000 KHz
1 Gigahertz(GHz)	= 1000 MHz

Note: Increase in the multiple of 10^3 i.e. multiple of 1000.

Accuracy *(शुद्धता। सटीकता)*

The computers are 100% accurate machine. The result obtained from the computer is extremely accurate and error free. Sometime it may produce undesirable result which is due to incorrect input. This term in computer is known as Garbage In Garbage Out (**GIGO**).

Automation *(स्वचालन)*

Computer is an automatic machine that is capable of functioning automatically according to the given predefined set of instructions in a program or software. During the time of data processing, the computer can automatically control the different devices that are attached to it to produce an output or result.

Versatility *(बहुमुखी)*

Since, the computer can perform various types of tasks, so it is a versatile machine. Its versatility makes it possible to be used in various fields like in education, healthcare, business and finance, entertainment, communication, engineering, defense, science and research, agriculture, government and public service, transportation, industries and manufacturing, etc.

Diligence *(लगनशीलता)*

A computer can perform complicated tasks repeatedly without getting tired and losing its speed, accuracy and efficiency so, it is a diligence machine. It never gets tired and never loses concentration while working.

Reliability (*भरपर्दो*)

A computer is reliable as it produces accurate output and is a diligent machine.

Storage (भण्डारण)

The computer can store huge amount of data, instructions and software in a small piece of hardware device. It also allows retrieving the stored information easily. The storage capacity of computer is measured in bit, byte, kilobyte, megabyte, gigabyte, terabyte, petabyte, etc.

Computer uses machine level language which is binary number using only two symbols 0 or 1. So, binary number system has a base of 2.

Symbols used in Binary Number System is known as **bit** which stands for "Binary Digit"

Single 0 or 1 is consider as One bit.

1 nibble	= 4 bit
1 byte	= 8 bit or 2 nibble or a character
1 Kilobyte(KB)	= 1024 byte
1 Megabyte(MB)	= 1024 KB
1 Gigabyte(GB)	= 1024 MB
1 Terabyte(TB)	= 1024 GB
1 Petabyte(PB)	= 1024 TB

Note: Increase in the multiple of 2^{10} i.e. in the multiple of 1024.

'b' stands for 'bit' where 'B' stands for 'Byte'.

Example: kb stands for kilobits, whereas, KB stands for Kilobytes.

Ability of Computer

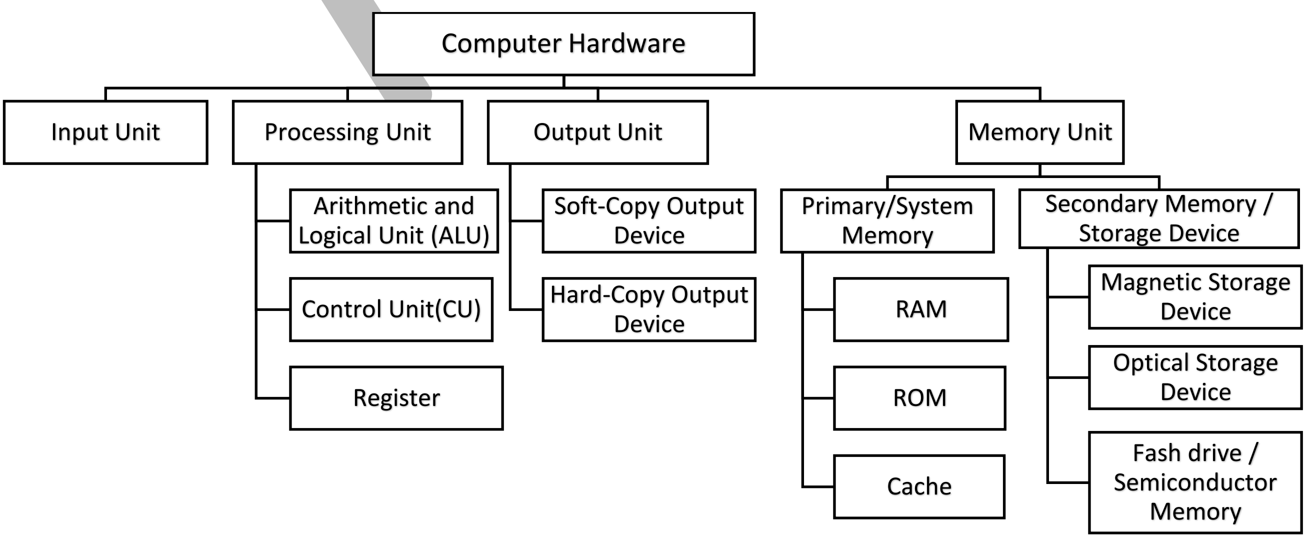
a. Accepts the data and instructions from the user.
b. Processes the data according to the given set of instructions.
c. Provides the result of processing to the user.
d. Stores the result permanently according to the requirement of the user.

Limitation of Computer

a. Computers lacks intelligence so, cannot think or make decisions on their own. They can only execute the instructions that they have been given.
b. Computers cannot understand or interpret complex social cues and emotions. They lack the ability to recognize, understand, and respond to subtle changes in tone, facial expression, and body language.
c. Computers cannot be creative. They cannot generate new ideas or concepts on their own.
d. Computers cannot feel emotions. They do not have the ability to experience or express emotions as humans do.
e. Computers cannot perform tasks that require physical dexterity or manipulation, such as assembling a piece of furniture or playing a musical instrument.
f. Computers cannot perform tasks that require common sense or practical knowledge, such as knowing how to drive a car or cook a meal.
g. Computers cannot perform tasks that require moral judgment or ethical decision-making. They do not have the ability to understand or consider the ethical implications of their actions.

Computer Hardware

All the physical, mechanical, tangible or electronics parts of a computer that can be touched, seen and felt.



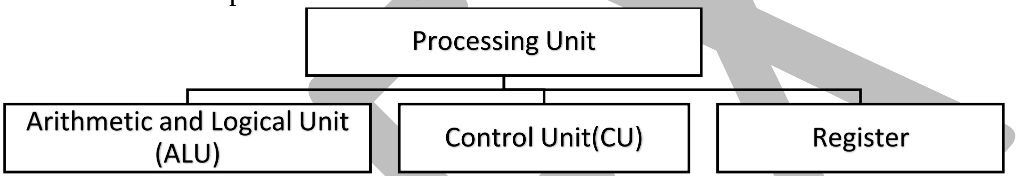
Input Unit

Input Unit consists of input devices that accept data and instructions or commands from the user and convert them into a suitable binary form that is accepted by the computer. It is also considered the **eyes** and **ears** of the computer. Input devices can be further grouped into the following types:

- **Primary input device**: Keyboard
- **Pointing devices**: is a hardware device that allows a user to control the movement of a cursor on a computer screen. It enables users to interact with graphical user interfaces (GUIs) by translating physical movements into digital inputs. Also, known as devices used in place of the mouse as the mouse is a commonly used pointing device. Some of them are: Mouse, Touchpad, Joystick, Trackball, Light Pen, Digitizer or graphics Tablet, Touch Screen, Stylus, Presenter, etc.
- **Scanning devices**: Scanner, Bar-code reader, QR (Quick Response) code reader, MICR (Magnetic Ink Character Recognition), OCR (Optical Character Reader), OMR (Optical Mark Reader), etc.
- **Audio input device**: Microphone,
- **Image and video capturing device**: Webcam, Digital Camera, etc.

Processing Unit

Processing unit also known as Central Processing unit (CPU) is considered as a brain of a computer. It performs all the operations of data according to program’s instructions. Like a human brain, this device also performs calculation, executes the program instructions and controls and coordinate other parts of the computer what to do. It consists of three main parts as follows:



a. Arithmetic and Logical Unit (ALU)	**b. Control Unit(CU)**	**c. Register**
• During the time of data processing the CPU has to perform different mathematical and logical operations on data according to the provided instructions and the ALU unit of CPU performs those arithmetic and logical operations. • ALU performs all the arithmetic (अंकगणित) (additional, subtraction, multiplication, division, etc) and logical (तार्किक) (AND, OR, NOT, comparison, etc) operation in computer.	• It fetches(ल्याउँछ) the instructions from the input device or memory unit, then decode and executes (कार्यान्वयन) the instruction. • It generates the control signals and control and coordinate all the parts and peripherals devices of the computer.	• Register एउटा सानो मेमोरी हो जसले डेटा निर्देशनहरू राख्छ जुन CU र ALU मा प्रशोधन भइरहेको छ। साथै, यो प्रोसेसर भित्रको सबैभन्दा सानो मेमोरी हो जुन प्रोसेसरको गतिमा काम गर्न सक्छ। • Register is the small amount of memory available inside processor responsible for holding data instruction while execution by CU and ALU. Also, it is the smallest memory inside a processor that can operate at the speed of the processor.

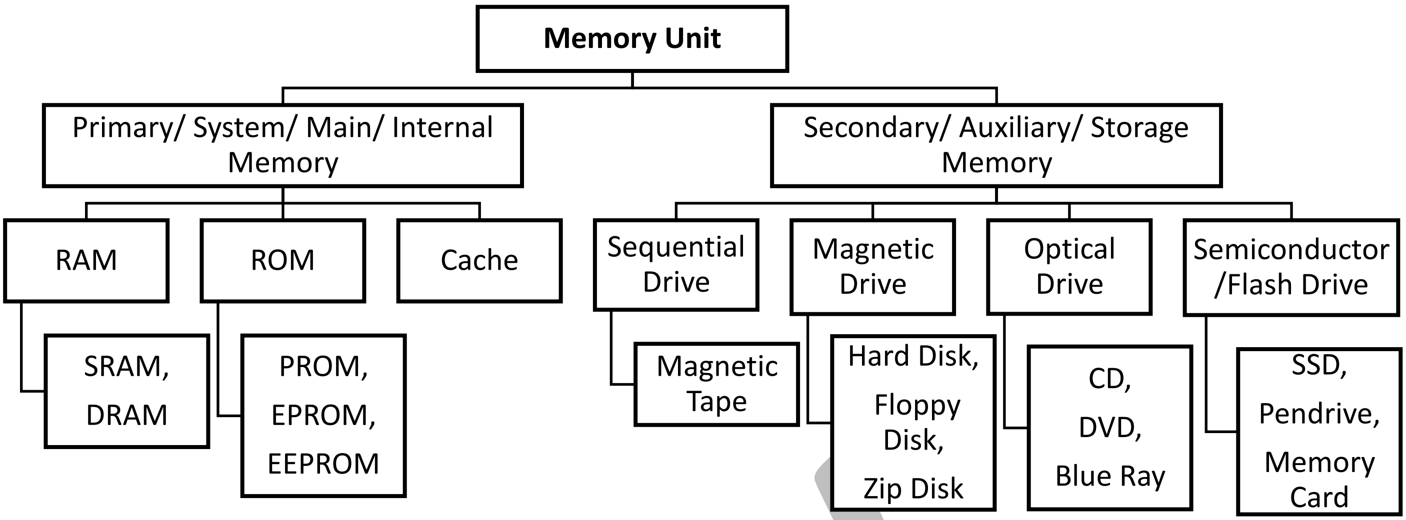
Output Unit

Output Unit consist of output devices that are responsible to provide information to an user after data processing. It converts produce information into suitable form that can be understood by an user. E.g.: Monitor, Printer, Speaker, Projector, Plotter, etc.

Memory Unit

Memory unit consist of memory devices that are used to store data, information and programs either permanently or temporarily. Memory unit can be further categorized into primary and secondary memory.

Primary Memory	**Secondary Memory**
• Also known as: system, main or internal memory पनि भनिन्छ। • data, programs र files हरू राख्छ जुन processing को समयमा भण्डारण गर्न आवश्यक हुन्छ। • It is the fastest memory compared to secondary memory. • They can be both temporary or permanent memory. (volatile or non-volatile) • Example: RAM, ROM and Cache	• Also known as Auxiliary memory or storage device • This memory is used to store data, information and program files permanently for future use so, also known as a storage unit. • It is slower compared to Primary Memory but can store a huge amount of files. • Are always permanent (non-volatile) • Example: Hard disk, pen drive, CD, etc.



Motherboard: Also, known as system or main circuit board that connects all components of a computer.

RAM (Random Access Memory): Temporary (Volatile) memory used to store data being used by the system. Also, known as working memory of the computer

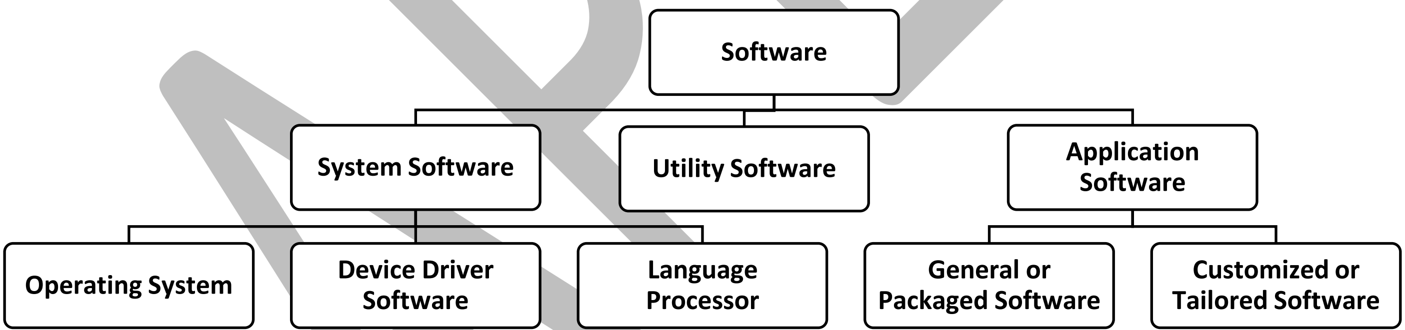
ROM (Read Only Memory): Memory that contains firmware(BIOS) which provides direction to startup the computer.

Cache Memory: High-speed buffer memory between CPU and RAM used for storing frequently accessed data.

Power Supply Unit (PSU): Provides electrical power to the different parts and component of computer system according to their requirement.

Computer Software

Software is the non-physical parts of a computer, made up of a collection of programs, that makes the computer hardware work. Users run various types of software on computer hardware to perform their tasks on the computer.



1. **System Software:** It is a hardware-oriented software that manages and supports the computer resources and operation of computer. Its types are:
   a. Operating System (OS): OS is system software which operates the computer hardware, provides user-interface and enables the application software to run on computer. Example: Windows 10, macOS, Linux, etc.
   b. Device driver: is a special kind of software or program that controls a specific hardware device attached to a computer like NIC, sound card, printer, graphics card, etc.
   c. Language processors: are special software used to translate computer languages. Example: Assembler, Complier and Interpreter.
2. **Utility Software:** is a collection of programs that help users manage and optimize their computer systems. It's designed to support the computer's infrastructure, and includes tasks such as:
   - Backing up and restoring data
   - Compressing files
   - Defragmenting the hard disk
   - Diagnosing hardware faults
   - Detecting viruses
   - Repairing file storage errors

3. **Application Software:** It is a user-oriented software designed to fulfill the specific purpose of a user to make their task easier. Through this type of software, a user performs the tasks they want or need on a computer.
   a. **Tailored or Customized Software**: यस प्रकारका सफ्टवेयरहरू संगठन वा व्यक्तिको विशिष्ट आवश्यकताहरू समाधान गर्न डिजाइन गरिएको हो। Example: Mobile banking apps, Information System software, etc
   b. **General or Packaged Software**: यस प्रकारका सफ्टवेयरहरू समग्र संगठनहरू र व्यक्तिहरूको सामान्य आवश्यकताहरू समाधान गर्न डिजाइन गरिएको हो। यी प्रकारका सफ्टवेयर सामान्यतया प्याकेजमा उपलब्ध हुन्छन् त्यसैले packaged software पनि भनिन्छ। Example: Microsoft Office with Word processor, Spreadsheet, Presentation, Google Office pack with google docs, sheets, slides and forms, Libre Office, etc.

Computer Network

कम्प्यूटर नेटवर्क भन्नाले दुई वा बढी कम्प्युटिङ् यन्त्रहरूको अन्तरसम्बन्धलाई जनाउँछ जसले डेटा आदानप्रदान गर्न र स्रोतहरू एकअर्कासँग साझेदारी गर्न सक्छ।

A **computer network** is defined as a system that connects two or more computing devices for transmitting information and sharing resources.

Communication Media

Communication media is a medium or channel through which computers on a computer network are connected with each other.

1. Wired or bounded or Guided Media
   - Twisted pair cable(Ethernet), Coaxial Cable, Optical fiber, etc
2. Wireless or unbounded or Unguided Media
   - Satellite, microware, radio wave, Wi-Fi, Bluetooth, Infrared, etc

Types of computer networks on the basic of coverage are:

1. LAN –Local Area Network
   - Covers small area like room, flat or a building.
   - Uses twisted pair cable(ethernet), wifi,
   - Example: office or home network, etc
2. MAN – Metropolitan Area Network
   - Covers small geographical area like town, city or a village.
   - Uses coaxial cable, optical fiber, microwave,
   - Example: CCTV network in a metropolitan city, network of ISP, etc
3. WAN – Wide Area Network
   - Covers a large geographical area like different countries or even continent.
   - Uses Submarine Optical Fiber connection, Satellite,
   - Example: Internet
4. PAN -Personal Area Network
   - Limited to small space between 1 to 10 meter or a body
   - Uses Bluetooth, Infrared,
   - Example: Wireless headset, Bluetooth speaker and other technology using Bluetooth

Internet

- It is a worldwide system of interconnected computer networks and electronic devices that communicate with each other using an established set of protocols.
- Internet is an example of Wide Area Network (WAN).
- Also, known as network of networks.
- It is a public computer network

Services of Internet

- **Communication Services:** Email, Telnet, Newsgroup, Instant messaging, VoIP, etc.
- **Information retrieval Service**: FTP, Search Engine, etc.
- **WWW (**World Wide Web**):** Manages websites its resources spread over the several servers over the internet.
- **Web Service:** Web Application and other cloud services.

इमेल (E-mail)

Electronic mail (**email** or e-mail) is a method of transmitting and receiving messages using electronic devices.
This is a service of the Internet, it is also called electronic postal service. With this service, we can send and receive our electronic letters from one place. That is why email services have become popular today.
Queen Elizabeth II sent the first email over the ARPANET from the Royal Signals and Radar Establishment in England on March 26, 1976.

इमेलका फाइदाहरू (Advantages of E-mail):

- Delivery of messages is fast and instant,
- The cost of e-mail is almost free or negligible.
- Multiple copies of the same message can be sent to a group of recipients at the same time,
- Images, documents, and other electronic files can also be attached to the e-mail.

Disadvantages of Email

- The slightest error in the email address or a failure in one of the links between the sender and the receiver fails email delivery.
- Although email is delivered instantly, the recipient may or may not read the mail on time.
- Since email passes through a network, it may be intercepted in between.
- Failure of electricity and telephone systems can prevent the user from sending or receiving emails.

Information and Communication Technology (ICT)

- The phrase "Information and Communication Technology" has been used by academic researcher since the 1980's and it became popular after it was used in a report to the UK government by Dennis Stevenson in 1997.
- It is technological tools and resources used to communicate, create, disseminate, store and manage information.
- Information can come in many forms such as sound, video, text, and images, so when you think of what technology is available that produces these aspects of information and sometimes a combination of all these, we refer to such technology as mobile phones, digital cameras, video cameras for example.
- Today information and communication technologies are one thing and so the repertoire of technologies expands further to encompass computers and computer-related products, email, MMS, and other forms of communication

- Broadly ICT is a combination of two major terms:

  a. **सूचना प्रविधि Information Technology(IT)**
    - It is concerned with the data processing, management, storage, and retrieval system, basically using computers and computer networks.
    - The Key task of a computer is to process data to generate information.

  b. **सञ्चार प्रविधि Communication technology**
    - It is mainly concerned with the efficient transmission or distribution of information.
    - It is use of different mean for communication like telephone, radio, television, wire, satellite, or cable. Computer network and internet is the main mean of communication technology used todays'.

- Due to the development of information and communication technology, the world has changed into a **Small Global Village**.

ICT Tools

- ICT tools are the devices, objects and necessary software used in information and communicating such information and communicating such information worldwide easily and effectively.
- Previously the major ICT tools were newspapers, radio, television, and telephones but due to the development in computer technology, today the major ICT tools include computers' hardware, software, network devices, and peripherals.
- Peripheral devices like: Camera, Scanner, printer, pen-drive , projector, microphone, etc.
- Application Software like: word-processing, spreadsheet, presentation, graphical, multimedia, etc.
- Network Devices like: NIC, switch, hub, router, repeater, gateway, etc.
- Devices like: Mobile phone, smartphone, PDA, etc.
- Technologies like: Computer Network, wifi, Internet , E-mail, Telnet, etc

Some important dates of the invention of ICT tools are:

- 1609 AD First printed newspapers, published weekly in Germany
- 1849 AD Invention of the photocopy machine
- 1876 AD Invention of Telephone
- 1895 AD Invention of Radio
- 1925 AD Invention of Television
- 1990 AD ARPANET was decommissioned and Internet was formed.
- 1990 AD HTML was invented by Tim Berners-Lee,
- 1991 AD 1st website was published on www by Tim Berners-Lee,
- 1992 AD Internet was made public and commercial data were allowed to be transported.

Multimedia

- Multimedia is a term formed by the combination of two words 'multi' and 'media'.
- Multi means many or multiple and media means the tools or medium used to represent or deliver information. As the name implies multimedia is the combination of different media.
- Multimedia has five different elements:
    - Text – लेख्य (basic, simplest and often the most effective way to express any information)
    - Graphics/ Image – दृश्य (non-text information such as drawing, charts, or photographs)
    - Audio / Sound – श्रव्य (form of speech, sound effects and also music score)
    - Video – भिडियो (refers to the pictures in motion.)
    - Animation – एनिमेसन (the illusion of motion created by the consecutive display of images)

Thus, multimedia is the field concerned with the computer-controlled integration of text, graphics, drawing, Video, animation, audio, and any other media where every type of information can be represented, stored, transmitted, and processed digitally.

Advantages/Benefits of using ICT in Education:

- Enhanced the modes of communication
- Cost-efficient
- Paperless: Eliminate the usage of paper. Eco-friendly
- Enhanced data and information security
- Minimize cost and save time
- Automatic solutions to manual paper-based processes and procedures
- Spread awareness of the social impact of technological change
- encourages collaboration
- Enhances subject learning:
- E-learning or Online Learning
- Independent learning platforms for students
- ICT use develops ICT literacy and ICT Capability
- Promotes higher-order thinking skills
- ICT brings inclusion
- Improve the administration and enhance the quality and efficiency of office work
- knowledge economy
- Promote and improve the digital culture in office work

Disadvantage of using ICT

- Misleading and misguiding information
- Risk of cyber-attacks and hacks
- A risk to the traditional book and handwriting methods
- Misuse of technology
- Not accessible everywhere
- Users require experience to handle ICT
- Implementing computers and the internet is expensive
- Few believe that computers can limit imagination

Challenges of ICT

- unstable internet connection,

- lack of training,
- Lack of adequate ICT resources (hardware and software),
- lack of materials and time,
- Culture, community mentality
- Cost of ICT resources
- Frequently changing technology

ध्यान दिनुपर्ने कुराहरु

- Tracking of Digital Footprint, Identity theft, hacking and cheating
- Bullying trolls, stalkers, and crimes,
- Spam and advertising,
- Addiction, time-waster, and causes distraction,
- Malware attacks,
- Pornographic and violent images,

Abbreviations used in ICT

WWW: Services of inter that manages websites its resources spread over the several servers over the internet.

Webpage: A web page is a single digital document on the World Wide Web, typically written in HTML, that can be viewed in a web browser. It contains information similarly to page in book but supports audio, video and animation along with text and graphics.

Website: is a collection of related web pages. Also consider as a book of webpages.

Web or website Address or Domain Name: A unique name (like www.example.com) used to identify and access a website on the internet.

Home Page: The main or introductory page of a website, typically serving as the starting point for navigation. Also known as index page.

Hyperlink or URL (Uniform Resource Locator): is a clickable link that directs to another resource or page, and a URL is the web address used to locate that resource.

Web Application: A software application that runs on a web server and is accessed through a web browser over the internet.

Browser: Application software used to surf internet. Example: Chrome, Edge, Firefox, internet explorer, Brave, etc.

Protocol: Set of Rule, TCP/IP, FTP, HTTP, SMTP, POP, IMAP, HTTPs, etc.

FTP (File Transfer Protocol): Used in Downloading and Uploading a file.

Downloading: A process of transferring files stored in servers to user’s computer though network like internet.

Uploading: A process of transferring files stored in user’s computer to server though network like internet.

Server: Powerful computer that provides resources, data, services, or programs to other computers, known as clients, over a network. Example: Data Server, Web Server, Mail Server, etc.

Search Engine: An internet tool that searches and retrieves information from the internet based on user-entered keywords or queries. Example: Google, Bing, Ask, etc.

IP Address: A unique identifier for a device on a network.

Router: A device that routes data from one network to another.

Modem: A device that converts digital signals to analog and vice versa.

Cloud Computing: Using remote servers to store and access data online.

VoIP (Voice over Internet Protocol): Making voice calls using the internet.

Bandwidth: The maximum rate of data transfer over a network.

Ethernet: A common method of wired network connection.

E-learning: Learning conducted via electronic media.

Teleconferencing: Real-time audio or video communication.

ERP (Enterprise Resource Planning): Business management software.

MIS (Management Information System): System for managing information in an organization.

Password: A secret and authorized set of characters that acts as a key to lock or unlock computers, data, and software.

Firewall: Security system that controls incoming and outgoing network traffic.

Antivirus: Software to detect and remove malicious software.

Encryption: Converting information into a secure format.

Digital Signature: Cryptographic method of verifying the authenticity of digital messages or documents.

VPN (Virtual Private Network): Creates a secure connection over a public network.

Digital Footprint: The trail of data you leave behind while using the internet.

Biometric Authentication: Security methods using physical characteristics like fingerprint or facial recognition.

Information Security: Protection of data from unauthorized access or corruption.

Phishing: A method of trying to trick users into giving personal information.

Cyber Space: कम्प्युटर र कम्प्युटर नेटवर्क उपकरणहरूको प्रयोग द्वारा सिर्जना गरिएको एक भर्चुअल स्पेस हो।

Cyber Crime: Illegal or criminal activity or electronic fraud committed by the people using ICT.

Password Guessing: Attacker हरूले words हरूको dictionary लिन्छन् र special software को प्रयोग गरी पासवर्ड अनुमान गर्न र बुझ्न प्रयास गर्छ।

Hacking: process of gaining unauthorized access to personal computers, networks, or information systems. (अनाधिकृत पहुँच प्राप्त गर्ने प्रक्रिया)

Cracking: Cracking is the process of breaking security system of copyrighted software or information systems so that the software or information can be used without paying royalties. (सफ्टवेयर वा सूचना प्रणालीको सुरक्षा प्रणाली तोड्ने प्रक्रिया)

Software Piracy: It is the act of illegally using, copying, or distributing software without the permission of the software company or vendor. सफ्टवेयर को गैरकानुनी रूपमा प्रयोग, प्रतिलिपि वा वितरण गर्ने कार्य।

Harassment: The perpetrator uses technology, such as a computer with access to the Internet or social media, or any other device, to harass, stalk, or abuse another person by participating in online hate campaigns.

Plagiarism: Act of incorporating someone else's work or ideas into your own work without their consent or full acknowledgement. साहित्यिक चोरी भन्नाले अरू कसैको काम वा विचारहरूलाई उनीहरूको सहमति बिना वा पूर्ण मान्यता विना आफ्नो काममा समावेश गरेर आफ्नो रूपमा प्रस्तुत गर्नु हो।

Phishing Attack: A type of social engineering attack that is often used to steal user’s confidential data, including login credentials and credit card numbers. (सामाजिक इन्जिनियरिङ आक्रमण)

Pornography: यो मानव यौन व्यवहारको प्रतिनिधित्व हो जुन इन्टरनेट मार्फत् कम्प्युटर प्रयोगकर्तालाई वितरण गरिन्छ।

Cyber Ethics: refers to the set of moral principles and guidelines that govern the use of computers and digital technology.

Cyber Hygiene: Cyber hygiene relates to the practices and precautions users take with the aim of keeping sensitive data organized, safe, and secure from theft and outside attacks. to maintain system health and improve online security of computing system and its resource.

Privacy: Respecting personal data and preventing unauthorized access.

Intellectual Property: Not copying or distributing software, music, or documents without permission.

Digital Responsibility: Avoiding cyberbullying, hacking, and misuse of data.

Professional Conduct: Following fair practices in IT professions.

Algorithm: A step-by-step set of instructions for solving a problem.

Database: A structured collection of data (e.g., MySQL, Oracle).

AI (Artificial Intelligence): Machines performing tasks that typically require human intelligence.

IoT (Internet of Things): refers to a network of physical objects such as devices, vehicles, appliances, and other items embedded with sensors, software, and connectivity that allow them to collect and exchange data over the internet without human intervention.

Blockchain: A decentralized ledger technology used in cryptocurrencies.

Big Data: Extremely large data sets that require advanced tools to analyze.

Machine Learning: A type of AI where systems learn from data, past experience and improve over time.

Booting: The process of starting up a computer and loading the operating system.

HTML (HyperText Markup Language): Standard language for creating web pages.

CSS (Cascading Style Sheets): Language used to style HTML content.

JavaScript: A scripting language used to make websites interactive.

Web Server: A server that hosts websites and delivers web pages to users.

Cookie: A small data file stored on a user's device by a website to remember preferences or login status.

CRM (Customer Relationship Management): Software for managing company interactions with customers.

ICT Infrastructure: The combined hardware, software, networks, and services required for ICT operations.

Business Intelligence (BI): Tools and systems that help analyze business data for decision-making.

EDI (Electronic Data Interchange): Exchange of business documents between companies electronically.

Telecommuting: Working remotely using ICT tools like email, video calls, and cloud services.

Virtual Reality (VR): Simulated experience using computer technology, often immersive.

Augmented Reality (AR): Overlays digital content onto the real-world environment.

Edge Computing: Processing data near the source of data generation instead of a central server.

5G: The fifth generation of mobile network technology, offering faster speeds and lower latency.

Digital Transformation: The integration of digital technology into all areas of a business.

Word processing: is the use of computer software to **create, edit, format, and print text documents**. It allows users to type, organize, and format written information with tools for font styles, margins, alignment, spell check, and more. Commonly used for Writing letters, reports, application, recommendation, CVs, resumes, and essays. File created is known as "document", contains pages and file has an extension of ".doc" or ".docx". Example: Microsoft Word, Google Docs, LibreOffice Writer, Word Star, Word Perfect, etc.

Spreadsheet: is a type of software that organizes data in rows and columns (cells) and is commonly used for data entry, storage, calculations, analysis, and presentation in various charts and diagrams. It includes built-in functions and formulas for mathematical, statistical and trigonometrical operations. Commonly used for Budget tracking, financial analysis, attendance records, inventory management. File created is known as "Workbook", contains sheets or worksheet and has an extension of ".xls" or ".xlsx". Example: MS Excel, Google Sheets, LibreOffice Calc, Lotus 1-2-3, WikiCalc, etc.

Presentation: is used to create and display slideshows consisting of text, images, audio, video, and animations for visual communication during lectures, business meetings, or seminars. Commonly used for Educational lectures, business pitches, training sessions, product demos, etc. File created is known as "presentation", contains "slide" and has an extension of ".ppt" or ".pptx". Example: MS PowerPoint, Google Slides, Impress, Presenter, Canva, etc.

Short Cut Keys

Ctrl + A	= Select All	Ctrl + M	= Tab(Forward Handing indent
Ctrl + B	= **Bold** (on/off)	Ctrl + N	= Create a new window or blank document
Ctrl + C	= Copy	Ctrl + O	= Open File
Ctrl + D	= **Display the Font Dialog Box**	Ctrl + P	= Print
Ctrl + E	= Alignment Center	Ctrl + R	= Alignment Right
Ctrl + F	= Find	Ctrl + S	= Save the document
Ctrl + G	= Goto	Ctrl + U	= Underline (on/off)
Ctrl + H	= Find and Replace	Ctrl + V	= Paste
Ctrl + I	= *Italic (on/off)*	Ctrl + W	= File Close
Ctrl + J	= Alignment justify	Ctrl + X	= Cut
Ctrl + K	= Hyperlink	Ctrl + Y	= Redo
Ctrl + L	= Alignment Left	Ctrl + Z	= Undo

Note: There is no shortcut key for Ctrl + T and Ctrl + Q

Ctrl + Shift + D	= Double Underline	Alt + F4	= Exit from current program
Ctrl + Shift + P	= Font Size	Ctrl + [	= Reduce font size by one digit
Ctrl + Shift + F	= Font Style	Ctrl +]	= Enlarge font size by one digit
Ctrl + Shift + (+)	= Superscript (e.g. 1^{st}) on/off	Ctrl + 1	= Single space (Line)
Ctrl + (+)	= Subscript (e.g. O_2)	Ctrl + 2	= Double space (Line)
Ctrl + Shift + >	= Enlarge font size by two digits.	Ctrl + 3	= 1 ½ space (Line)
Ctrl + Shift + <	= Reduce font size by two digits.	Ctrl + Enter	= Hard Page Break

Some of the Short-cut key used in PowerPoint

- **"F5"** key is used to start the slide show view from the beginning whereas the "**ESC"** key is used to terminate the view in middle.
- "**Shift + F5**" key is used to start the slide show view from the current slide.
- "**B**" key is used to make the screen black and "**W**" key is used to make the screen white during the slide show.
- "**Ctrl + D**" is used to duplicate the slide.
- "**Ctrl +M**" is used to insert the new slide.

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